

Designs and Materials of Electrodes for Electrochemical Sensors

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Electrode material selection and structural designs of electrochemical chips are fundamental parameters in the field of electrochemical sensing. These parameters directly affect sensor conductivity, selectivity, stability, surface area, and overall performance. This article summarizes the most common electrode architectures and commercially available materials currently used in the development of electrochemical sensors, including carbon-based materials (e.g., boron-doped diamond, graphite, graphene, glassy carbon, carbon nanotubes, and carbon fibers), metal-based

materials and alloys (e.g., gold, platinum, silver, nickel, and metal oxides), conductive polymers (e.g., polyaniline, polypyrrole, and poly(3,4-ethylenedioxythiophene)), and redox dyes and mediators (Prussian blue, Meldola blue, etc.). It highlights the advantages of each category and identifies suitable electrode materials for specific target analytes. Finally, this review aims to guide readers in selecting appropriate electrode materials and designs tailored to a specific application.

1. Introduction

Enhancing the quality of life is one of the primary objectives of scientific research worldwide. To achieve this goal, techniques and methods must be developed to enable real-time monitoring, rapid diagnosis of prevalent diseases, ensure food safety, and support environmental monitoring. At the same time, maintaining the affordability of these technologies is another critical factor that requires careful consideration.^[1]

Several studies have reported the development of analytical methods that offer rapid, accurate, and selective responses.^[2–4] The available approaches include various detection methods, including optical techniques such as spectrophotometry and fluorometry, as well as piezoelectric, magnetoelectric, and electrochemical methods.^[5] Particularly, electrochemical methods play a significant role by offering versatile and sensitive approaches based on different detection methods including voltammetry, amperometry, potentiometry, and electrical

impedance spectroscopy.^[6] These methods typically use a classical electrochemical setup or more advanced configurations based on the use of electrochemical chips that contain all the electrochemical setup on the same support.

Nowadays, there has been growing interest in integrating recognition elements (nanozymes, enzymes, bacteria, viruses, DNA, antibodies, etc.) with electronic components to develop electrochemical sensors and biosensors for biological applications.^[7] These devices have demonstrated high effectiveness in identifying chemical and biological analytes by monitoring variations in electrochemical signals at the electrode interface. The key advantages of these devices include the potential for miniaturization and portability, along with high sensitivity and selectivity. They also offer a broad linear detection range, require minimal space and power, and are generally cost-effective in terms of instrumentation.^[8]

In electrochemical sensor and (bio)sensors, the materials used significantly affect the performance, the kinetics, and the efficiency of the electrochemical processes.^[9] The choice of electrode materials and design all play crucial role in determining the reaction rates, cell potential, and overall system stability.^[10] An ideal material for electrochemical applications must possess high electrical conductivity to facilitate efficient electron transfer between the electrode and the analyte(s) or reactant(s). Electrochemical activity is another essential property because materials should exhibit strong catalytic behavior or redox activity depending on the intended application (whether it is electrocatalysis or sensing).^[11] A relatively high surface area is often required, as it provides active sites available for electrochemical reactions to occur, thereby enhancing sensitivity and performance. Additionally, chemical and thermal stability are crucial to ensure that the material can support the temperatures and harsh reaction conditions without degrading or losing functionality over time. Selectivity is important in applications such as sensors or biosensors in order

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to distinguish between target molecules and interfering substances.^[12] Finally, for biomedical or environmental applications, biocompatibility is necessary, allowing materials to interact safely with biological systems or ecosystems.^[13]

Many forms of materials are available today for electrochemical applications, such as nanoparticles, thin films, nanowires, and printable inks. These materials can be integrated onto the electrochemical cells via different techniques including electro-deposition, physical vapor deposition (PVD),^[14] chemical vapor

deposition (CVD), and printing technologies (including screen printed,^[15] inkjet printing,^[16] and 3D printing).^[17]

On the other hand, the design of electrochemical cell is also of high importance. Key configuration aspects such as the number of electrodes, the substrate material, the surface area of each electrode according to its function, and the detection strategy (single mode detection, multiparametric detection, etc.) play all significant roles in determining detection efficiency. This review article aims to summarize recent advances in



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electrochemical cell design and materials used for electrochemical (bio)sensing applications.

2. Design of Electrodes in Electrochemical Sensory

To design electrodes for electrochemical sensors, several parameters must be considered followed by selecting the appropriate electrode materials. In this part, we describe the key parameters to consider when designing the electrochemical sensors and (bio) sensors (Figure 1). These include the number of electrodes, the substrate material, the surface area of each electrode according to its function, the detection strategy (single mode detection, multiparametric detection, etc.), and the manufacturing process with a particular focus on miniaturized systems such as screen-printed electrodes (SPEs) (Figure 1b).

Regarding the format of electrodes in electrochemical sensors, it can vary significantly depending on the application and the performance requirements. Common formats include planar electrodes, various types of printed electrodes such as 3D-printed,^[18] inkjet-printed,^[16] and SPEs,^[19] laser-engraved,^[20] microelectrode arrays,^[21,22] interdigitated electrodes,^[23] and

needle-based electrodes.^[24] Typically, conventional electrochemical cell configurations involve a two- or three-electrode systems as described at the beginning of this section (WE, CE, and RE) all set in standard (classical) electrochemical cells. These cells conventionally have volume of tens of milliliters adjusted according to the sample's volume and the intended experiment (Figure 1a). We can also distinguish between classical cells and thermostatic cells, which allow temperature control of the electrolyte using a jacketed setup connected to a water circulation system.

In this section, we will focus on miniaturized systems, particularly SPEs, as they represent a widely adopted, versatile, and cost-effective platform for both research and commercial electrochemical sensing applications. These electrodes are integrated on the same substrate, offering miniaturized and portable systems (Figure 1b). Common SPE configurations use carbon-based inks for CEs and Ag/AgCl for Res, whereas different in functionalized materials (ink-based) are commonly used for WEs and will be the subject of the next part of the article. SPEs typically use specific cells and may be also integrated to a microfluidic setup. The latter allows experiments to be performed under flow conditions which are more suitable for specific applications (organ-on-a-chip, lab-on-a-chip, etc.) (Figure 1c).^[6,25–29]

It is important to highlight that an analyte can be detected using a single working electrode (WE) (Figure 2a,b,c,d,i,k).

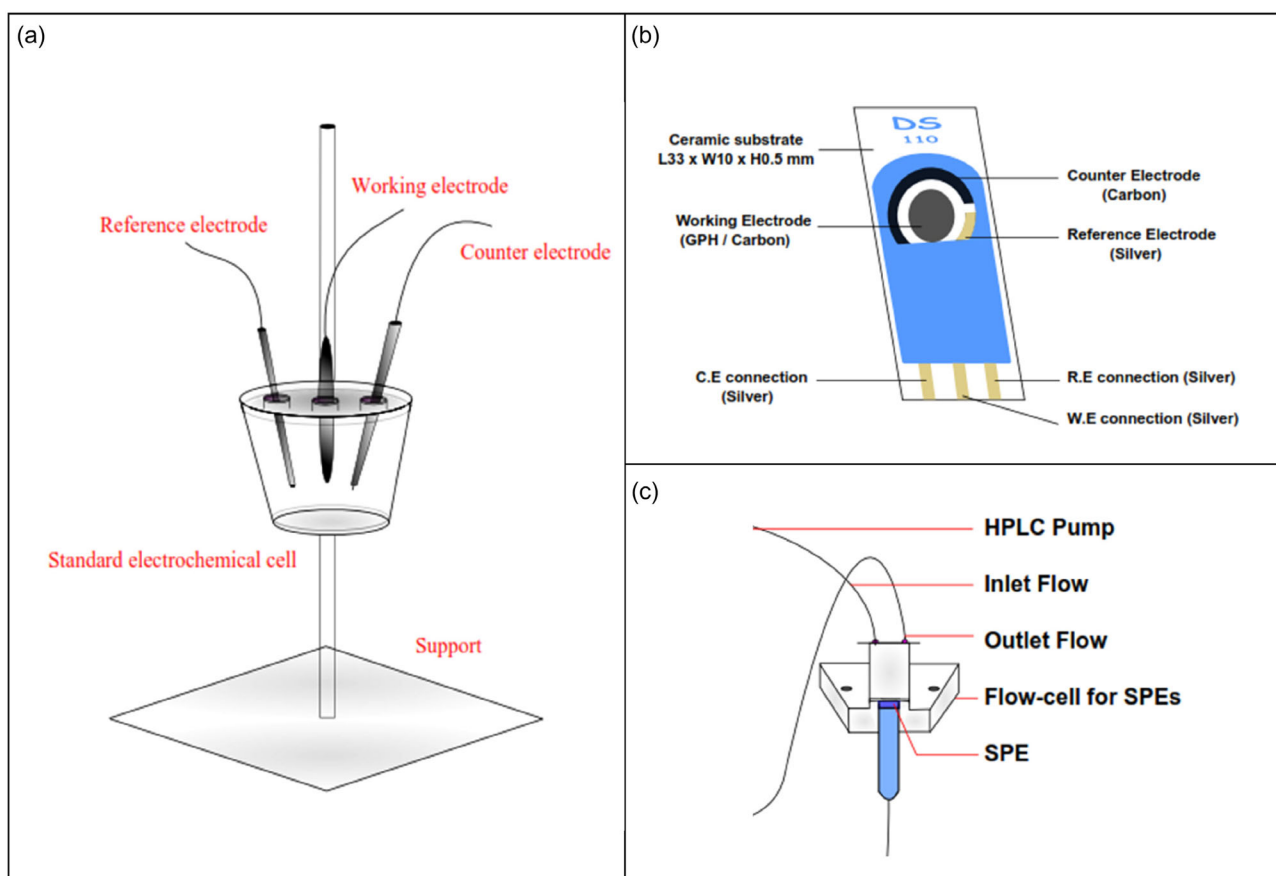


Figure 1. a) Classical electrochemical cell of three-electrodes setup including the working, counter and REs^[42]. b) Distribution of all electrodes on the chip of a SPE^[28]. c) Microfluidic cell (namely also flow cell) integrating a SPE to perform experiment under flow conditions.^[28]

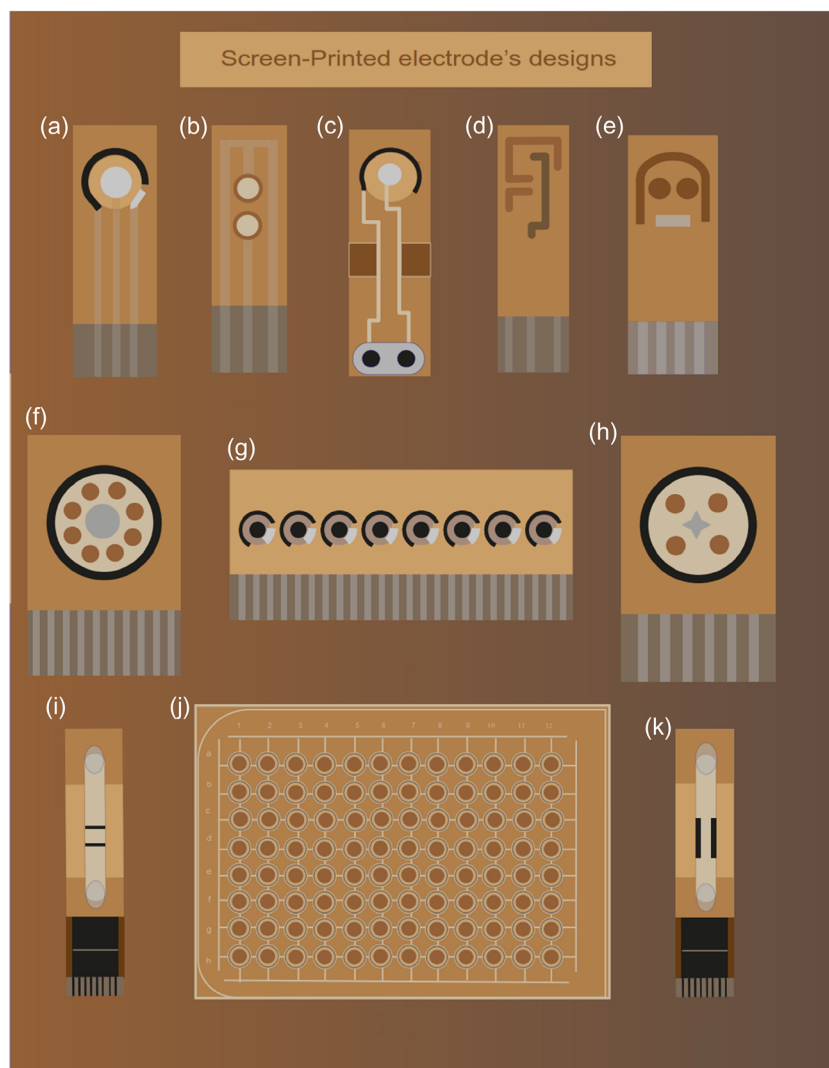


Figure 2. Common designs of SPEs. a) Classical design of a three-electrode system consisting of a WE (planar, positioned at the center, in gray), a CE (on the left, black), and a RE (on the right, in gray). b) Design of a two-electrode system consisting of electrode serving as a WE and the other one serving as a counter and RE at the same time. c) Design of a two-electrode system with special connectors using crocodile clips often used for potentiometric measurements, similar to design b but with different connection way. d) Specific design of a three-electrode system consisting of a WE (positioned at the center, in gray), a CE (on the left, semi-rectangle, in brown), and a RE (on the top, open-rectangle, in brown). e) Specific design consisting of dual WEs integrated within the same electrochemical cell (two planar electrodes positioned at the center, in brown), a CE (at the top, surrounding the working and REs, in brown), and a RE (rectangle, at the bottom, in gray). f) Design of a multiparametric platform or multielectrode array consisting of eight WEs (planar, in gray), a CE (at the center, surrounding the WEs, in black), and a RE (planar, at the center, in gray). g) Design of an electrochemical platform with eight independent electrochemical cells, each cell consisting of its own working, counter, and REs (typical design to perform measurements). h) Multielectrode array consisting of four WEs (planar, in brown), a CE (at the center, surrounding the WEs, in black), and a RE (star, at the center, in gray). i, k) Design of an electrochemical sensors integrated into a microfluidic chip for real-time monitoring of analyte under flow condition, designed as two-electrode system consisting of a WE and a reference and CE at the same time (electrode in black, positioned in (i) perpendicular to the flow and in (k) parallel to the flow). j) Design of a petri dish-embed electrochemical sensors for the monitoring of culture media environment during cell culture; electrodes are integrated at the bottom of the cell-culture wells, WEs (in black), and counter and RE at the same time (around the WEs, in gray)).

This configuration refers to a setup where only one WE is responsible for the electrochemical reaction under investigation. In contrast, it is possible to use multiple WE systems (up to 96 electrodes) where multiple electrodes may be used for the electrochemical investigation. These electrodes may be functionalized with the same sensing (bio)material to detect the same analyte.^[30] This approach can improve the reliability and accuracy of measurements by combining the results obtained with all the electrodes exposed to the same medium. Such designs are usually controlled using multiplexing techniques to manage and

process the electrochemical signals efficiently.^[31] Additionally, electrochemical chips can be designed for multiparametric detection, enabling the simultaneous detection of multiple analytes.^[32] In this configuration, many WEs are integrated onto the same electrochemical chip (Figure 2e,f,g,h,j) and functionalized with different sensing (bio)materials, each tailored to selectively detect a specific analyte of interest.^[33,34] This setup requires a sophisticated electronic interface capable of applying different potentials and executing different electrochemical techniques across channels.^[35,36]

Electrochemical sensor configurations typically involve two categories: 1) two-electrode system and 2) three-electrode system.^[6,25] Each design has its own advantages and limitations and is selected based on the intended application and measurement requirements. In the two-electrode system, the WE serves as the site of the electrochemical reaction, where the counter electrode (CE) that completes the circuit also serves as a pseudo-reference electrode (RE) (Figure 2a,b). In this configuration, the CE helps in fixing the applied potential difference.^[26] The main limitation of this system lies in its poor potential control where it is difficult in absence of true RE to maintain stable potential over time.^[27] As a result, it is used in simpler systems, mainly as a single-use or disposable low-cost sensors. The three-electrode system consists of three independent electrodes: the WE, the CE, and the RE that maintains a stable and well-known potential (e.g., Ag/AgCl or calomel).^[28,29] This system is more appropriate for applications that require stable potentials over time making it ideal for sensitive and accurate electrochemical investigations. This system typically includes a dedicated CE typically made of inert or conductive materials such as platinum or carbon. Furthermore, the geometry and surface area of electrodes impact current distribution and mass transport within the cell.

From a theoretical standpoint, the choice of electrode materials and surface structure significantly affects electron transfer kinetics and overpotential.^[37] Thus, it is highly recommended that the surface area of the CE be at least three times larger than that of a single WE or the sum of all WEs in a multielectrode configuration. This ensures uniform current distribution and minimize potential drop across electrochemical cell,^[38,39] helps to maintain a stable electrochemical environment, and ensures accurate control of the WE potential.^[40] Conversely, high surface area electrodes can increase capacitive background, potentially obscuring faradic signals. Thus, the WE area should be selected based on the experiment purpose. For kinetic studies, the area typically ranges from 10^{-4} to 10^{-6} cm² corresponding to electrode diameters of tens of μ m (microelectrodes). For electroanalytical studies, larger surface area electrodes are commonly used, ranging from 0.01 to 0.1 cm² corresponding to electrode diameters of ≈ 1 –4 mm (small disk electrodes). In contrast, the RE does not require a large surface area since its primary role is to provide stable and accurate potentials.^[5] Regarding the distance between electrodes, the separation between RE and CE is generally not critical; however, the distance between the WE and RE separation should be kept small typically between 1–3 mm to minimize the ohmic drop. On the other hand, the WE–CE separation should be larger (5–15 mm) to allow proper current flow and to avoid interference from the CE reactions.^[41] Finally, some applications require miniaturization in order to integrate the entire electrochemical cell within a specific geometric area (e.g., for portable or point-of-care devices). This step must consider the previously mentioned requirements to ensure the proper design of the electrochemical chip.^[16,18–24]

SPEs have rapidly emerged since their introduction to the market generating growing interest in the development of electrochemical (bio)sensor platforms. These devices incorporate all the electrochemical cell including the necessary electrodes (WE, CE, and RE) onto the same electrochemical chip

(Figure 1b and Figure 2). Their ease of mass production and customization, low cost, and availability make them ideal, accurate, and cost-effective tools for (bio)sensor development.^[42] In addition, the straightforward integration of these sensors into microfluidic systems (microfluidic cells) enables real-time monitoring of target species under controlled microfluidic conditions (Figure 1c and Figure 2i,k). These systems are typically used for the development of lab-on-chip and point-of-care devices.^[43] Regarding the substrate material of the electrochemical chip, solid materials such as silicon, glass, plastic, and ceramic are typically used for classical application. However, flexible substrate materials such as Kapton (polyimide), hydrogel, polyethylene terephthalate (PET), polydimethylsiloxane (PDMS), and paper-based and textile-based materials are typically used for application that require flexibility and relatively stretchable properties such as wearables (bio)sensors.^[44,45] Some materials are optically transparent or relatively transparent; these are particularly interesting for devices that combine electrochemical and optical sensing.^[46] Glass, indium tin oxide (ITO), and PET materials are typically used in applications that require relative transparency combined with the needed electrochemical properties.

In addition to SPEs, 3D-printed electrodes are gaining attention as customizable and adaptable alternatives. Using additive manufacturing, electrodes can be fabricated with tailored geometries, controlled porosity, and material-specific properties. This allows for a rapid prototyping and integration into wearable and point-of-care devices.^[47] Likewise, laser-engraved electrodes provide a fast, mask- and solvent-free method to create highly porous conductive carbon-based structures on flexible substrate such as polyimide.^[20,48] These emerging technologies enable progress across diverse applications particularly in portable, miniaturized, and flexible sensing platforms.

Finally, some applications have specific requirements that can be challenging to meet. For example, organ-on-a-chip applications equipped with electrochemical (bio)sensors that come in contact with culture media require a sterilization step prior to integration or before the experiment starts.^[49,50] This implies that the materials used in the construction of the electrochemical chip must support the sterilization conditions, which can be sometimes challenging and may affect the material's properties. Another important consideration is the biocompatibility of the materials used. In certain applications, the electrochemical cell may be in direct contact with the target system (e.g., cells, organs, organoids, and interstitial fluid).^[51] This implies that the materials do not induce any adverse effects or cytotoxic responses.^[52] So, according to the target application, a specifications list should be considered. To summarize, the previously mentioned requirements and conditions must be applied and respected to ensure proper design of the electrochemical chip and to obtain optimal electrochemical responses.

3. Materials of Electrode for Electrochemical Sensors

After the selection of the electrode design, each electrode material should be chosen independently to assess its electrochemical

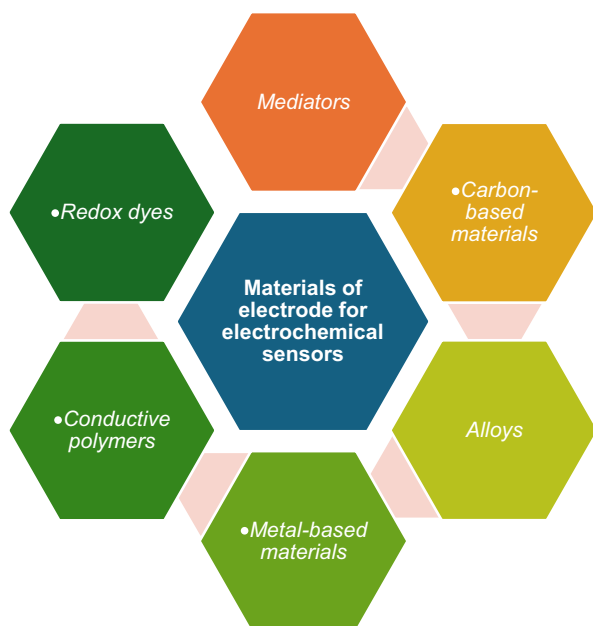


Figure 3. Common materials of electrodes used for the development of electrochemical sensors and (bio)sensors.

performance. Depending on the fabrication technology used, electrodes can be produced in small batches suitable for limited quantities or in large batches which reduce the production costs and enhance reproducibility. Additionally, the fabrication process may involve from three to dozens of steps depending on the method chosen (e.g., printing technologies, electrodeposition, and vapor deposition). It is important to highlight that each electrode material should be selected based on its specific function. For instance, the material of the RE is completely different from that of the WE. The RE differs fundamentally from the WE in its function and design. While the WE is the site of the redox reaction, the RE provides a stable and well-defined potential against which the WE potential is measured.^[53] RE typically does not pass current under normal conditions and is made of Ag/AgCl or saturated calomel, whereas the WE material varies depending on the target analyte (carbon-based, metal-based, dyes, and mediators, etc.). In this section, the materials used for the sensing electrodes are explored in detail to facilitate the selection of the most suitable material for the intended application (Figure 3). The materials are distinguished into two categories: primary sensing materials (such as carbon-based electrodes and polymer-based sensors) and modifying materials (such as metallic nanoparticles and metal oxides).

3.1. Primary Sensing Materials

3.1.1. Carbon-Based Materials

Carbon-based materials are widely used in electrochemical sensors due to their chemical stability, broad potential window, low background current, and analytical versatility. Unlike metal electrodes, they offer longer lifespans and better performance.^[54]

Nanostructured forms like graphene, graphite, and carbon nanotubes (CNTs) exhibit superior electrochemical properties, enabling highly sensitive and selective sensing, even in harsh conditions and across wide temperature and dynamic ranges. Meanwhile, boron-doped diamond (BDD) material exhibits a large potential window, making it a suitable for the detection of analytes at high potentials without interference from solvent oxidation/reduction peaks. Graphene and CNTs are distinct carbon polymorphs that have been extensively utilized as electrode materials in electrochemical sensing applications in recent years. Moreover, at room temperature, these carbon nanostructures offer high electrical conductivity, exceptional electron mobility, and a large surface area-to-volume ratio, further enhancing their performance in sensing applications.^[55] Each carbon-based material will be explored separately in this section to highlight its strengths and limitations (Figure 4).

3.1.1.1. BDD

BDD has attracted attention due to its remarkable electrochemical properties. It is mainly synthesized using microwave plasma-assisted chemical vapor deposition (MPCVD) at relatively high temperatures, with methane and hydrogen in the presence of diborane as the boron source.^[56] Alternatives for diborane such as trimethylboron (TMB)^[57] and boron trifluoride (BF₃)^[58] have also been used, offering good doping efficiencies. While microwave-assisted CVD remains the dominant method for synthesizing high-quality BDD, alternative techniques such as hot-filament CVD and DC arc discharge have been also reported.^[56] These methods offer varying trade-offs in terms of film uniformity, doping control, and scalability. Under these conditions, sp³ carbon hybridization predominates, while sp² content remains low.^[59] The sp³/sp² ratio of BDD films can significantly affect the sensing performance; for example, higher levels of sp²-hybridized carbon in BDD are associated with faster electron transfer, an increased background current, and enhanced susceptibility to corrosion,^[60,61] oxygen reduction,^[62] and wastewater treatment.^[63] Boron promotes diamond nucleation during growth and is associated with materials that exhibit smaller crystallite sizes.^[64]

Compared to many conventional electrode materials, such as platinum, gold, or graphitic carbon, BDD exhibits similar properties but with notable advantages. BDD is a highly promising electrode material for electrochemical sensor development. It offers exceptional electrochemical and physicochemical properties including a wide electrochemical potential window in aqueous solutions (3 V), high tolerance to extreme potential, high hardness, excellent wear resistance, very low oxygen sensitivity, and enhanced resistance to fouling compared to conventional materials, allowing access to a broader range of redox species and very low background currents.^[65–67] The capacitive current measured within this electrochemical window is significantly lower. In fact, the observed capacitive current is a few nanoamperes per square centimeter (nA cm^{−2}) on BDD, compared to several tens or even hundreds of microamperes per square centimeter (μA cm^{−2}) on conventional materials like platinum (Pt)

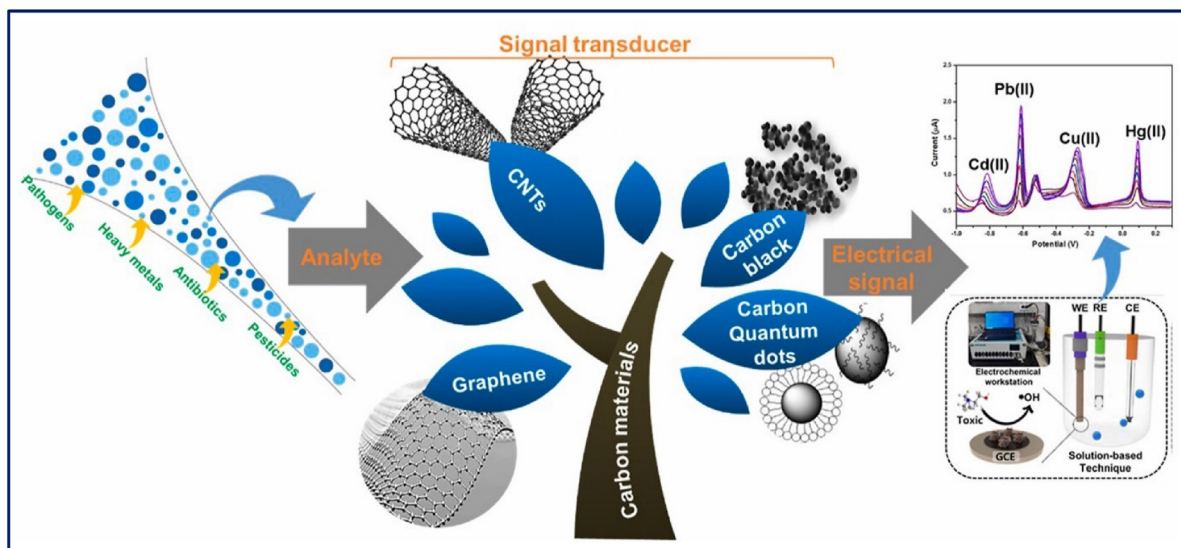


Figure 4. Common carbon-based materials used for the development of electrochemical sensors used in various applications including pathogens detection, heavy metals analysis, and antibiotics and pesticides monitoring.^[317]

and glassy carbon (GC) respectively. The absence of parasitic reactions on BDD reduces interference with the measurement of target analytes, thereby enhancing selectivity.^[61] In addition, BDD has been reported to exhibit excellent resistance to surface fouling which makes it a highly suitable candidate for biosensor development particularly in complex biological or environment matrices. This property is mainly due to its chemically inert surface and tunable termination which together reduce nonspecific adsorption on biomolecules.^[68]

An important aspect affects the performance of BDD electrodes is the activation procedure applied to the BDD surfaces.^[69] Particularly, anodic and cathodic preconditioning significantly affect the electrochemical behavior. Studies show that current densities associated with the electrochemical pretreatment of BDD electrodes alter the electrochemical response of BDD electrodes.^[70] Cathodic preconditioning generally results in hydrogen-terminated BDD surfaces, which are more hydrophobic and exhibit enhanced electrode transfer kinetics.^[71] However, anodic activation produces oxygen-terminated surfaces, more hydrophilic having slower electrode transfer profile, and low current-background.^[70,72] Thus, the choice of activation procedure impacts the BDD electrode sensitivity, selectivity, and stability. BDD electrodes have been used for wastewater treatment, being capable of generating hydroxyl radicals to degrade organic micropollutants.^[73] They are also useful for other applications such as CO₂ reduction, soil remediation, and electrogenerated chemiluminescence.^[65] BDD electrodes have been used for the quantification of the cytostatic drug flutamide in pharmaceutical formulations, spiked human urine, and water samples; the electrodes have also been used to detect atrazine in spiked river water samples.^[74,75] BDD-based sensing is useful for detecting pesticides, mycotoxins, peroxides, and phenolic compounds^[65] and inorganic chemicals (e.g., H₂O₂ and arsenic).^[76,77]

While BDD electrodes are widely used and effective, they present certain limitations that hinder their large-scale application. First of all, their synthesis presents a significant limitation: the columnar growth process requires several hours at a high temperature (>400 °C), and the resulting films are thick (several hundred nanometers), resistive, and rough contributing to bio-fouling.^[78] The high synthesis temperature limits the choice of deposition substrates, with silicon being the most commonly used substrate.^[79] Moreover, depositing BDD on more conductive and cost-effective supports (such as metallic alloys) or even integrating it as part of microelectronic processes is unfeasible. Except through complex transfer techniques,^[80] it is impossible to integrate BDD on flexible and low-temperature-resistant substrates.

3.1.1.2. Graphite Carbon

Pencil graphite electrodes (PGEs) have recently emerged as promising alternatives to conventional carbon-based and metallic electrodes in electrochemical applications. Primarily composed of graphite which is the most thermodynamically stable allotrope of carbon, commonly sourced from mineral ores such as coal,^[81] PGEs offer a unique combination of desirable properties. These include mechanical robustness, chemical inertness, low background current, a wide electrochemical potential window, strong analyte adsorption capacity, and excellent electrical conductivity due to the sp²-hybridized carbon network.^[82,83] Additionally, PGEs are inexpensive, disposable, and easily modified or miniaturized, making them particularly attractive for a variety of sensing platforms.

One of the most significant advantages of PGEs over traditional electrodes, such as GC electrodes (GCEs), is their renewable surface. Unlike GCEs, which require labor-intensive polishing

procedures, PGEs can be refreshed quickly and reproducibly, thus enhancing their utility in analytical applications.^[82,84] PGEs and graphite-modified electrodes have been effectively used in the voltammetric detection of a wide range of analytes, including genotoxic nitro compounds,^[85] organic pollutants,^[86] antibiotics,^[87] and heavy metals.^[88] Furthermore, preanodized PGEs (denoted as PGE*) have been successfully developed and utilized as renewable electrochemical sensors for detecting total phenolic preservatives in pharmaceutical insulin formulations,^[89] dihydroxybenzene isomers in tea dust,^[90] and purine metabolites such as hypoxanthine, xanthine, and uric acid in fish samples for food quality monitoring.^[4] Extensive comparative studies confirm the superior or comparable electrochemical performance of PGEs relative to traditional carbon electrodes, while screen-printed carbon electrodes (SPCEs) are often used as alternatives due to their low cost and ease of use. They are typically single-use and lack a renewable surface, which can limit reproducibility and long-term utility.^[91] In contrast, PGEs fabricated from commercially available pencil leads offer a more economical and sustainable option.

SPCEs have been extensively utilized in the development of electrochemical biosensors for diverse analytical applications, including the evaluation of nucleic acid damage,^[92] the assessment of antioxidant activity in beverages such as beer, coffee, and tea,^[93] and the simultaneous quantification of methocarbamol and paracetamol.^[94] Despite their widespread use, SPCEs are predominantly single-use, disposable devices with nonrenewable surfaces, which limits their reusability and long-term applicability.^[91] In contrast, PGEs, fabricated from commercially available pencil leads, present a significantly more economical alternative to SPCEs, while maintaining adequate analytical performance.

The material composition of pencil leads further supports their use as electrodes. Contrary to their name, pencil leads contain no metallic lead; instead, they are composed of graphite intercalated with clay and other binders.^[95] While high graphite content generally correlates with increased conductivity, the optimal electrode performance often depends on the specific interaction between the analyte and the graphite/clay matrix. For paper-based sensor configurations, factors such as the ease of depositing graphite material onto cellulose substrates also influence the selection of pencil hardness.^[96] In recent years, significant progress has been made in enhancing the functionality of PGEs through chemical modification. These modifications include the incorporation of metal and metal oxide nanoparticles, metal complexes, carbon nanostructures, and conductive polymers (CPs). Akanda et al. provided a comprehensive overview of these developments, highlighting the expanding role of PGEs in the design of chemical and biosensors.^[97] Torrinha et al. also reviewed enzyme-modified PGEs, though their discussion was limited to specific nanomaterial enzyme conjugates.^[98] Among these advancements, enzyme immobilization onto PGEs has received growing attention for the development of highly sensitive electrochemical biosensors. Since the late 2000s, a wide variety of enzymes including glucose oxidase, alkaline phosphatase, xanthine oxidase, cholesterol oxidase, horseradish peroxidase, urease, alcohol dehydrogenase, laccase, uricase, ascorbic oxidase,

lipase, glycerol kinase, and glycerol-3-phosphate oxidase, have been immobilized onto PGEs. These enzyme-modified PGEs biosensors offer an attractive combination of low cost, ease of fabrication, and robust analytical performance for applications in clinical diagnostics, food safety monitoring, and environmental analysis.^[83,91,99–101]

The continued development of effective enzyme immobilization strategies such as physical adsorption, covalent bonding, and entrapment in polymeric matrices has further enhanced the stability and reusability of PGE-based electrochemical biosensors. As a result, PGEs are increasingly recognized as a promising platform for next-generation biosensors, capable of replacing more conventional carbon electrodes in a wide range of analytical applications.^[102]

3.1.1.3. Graphene Carbon

Graphene, a prominent 2D carbon-based nanomaterial renowned for its exceptional electronic and mechanical properties, holds immense potential across a wide range of applications, including conductive wiring, energy storage and conversion systems, actuators, and catalytic processes.^[102–105] Graphene and its related materials currently represent the cutting edge of high-performance carbon technologies^[106] as evidenced by the European Union Research Council's significant initiative known as the EU Graphene Flagship.

2D graphene has long been explored as a versatile material platform, owing to its flexible structure and tunable surface chemistry, for the development of high-performance electrocatalysts. In recent years, a growing array of graphene derivatives and analogues has continued to emerge, demonstrating remarkable potential for highly active and selective electrocatalytic applications. The progression of studies in this field began with the fabrication of wearable sensors using reduced graphene oxide (GO) patterned on paper substrates.^[107] This was followed by the development of porous, coral-shaped molybdenum carbide-graphene composites used as piezoresistive sensing materials, produced via direct laser writing on paper substrates.^[108] Most recently, research has advanced to the creation of graphene-based flexible and wearable strain sensors.^[109] Sekula-Stryjewska et al.^[110] investigated both GO and reduced GO (rGO) as facilitators of cardiomyogenic and angiogenic differentiation in human mesenchymal stem cells under *in vitro* conditions. Ginestra et al.^[111] developed electrospun composites of poly(caprolactone) integrated with graphene for applications in nerve tissue engineering. In a similar vein, Polo et al.^[112] engineered nanostructured scaffolds composed of poly(lactic acid)-poly(caprolactone) copolymers incorporating GO, with the objective of promoting aligned cell migration and enhancing the neuronal differentiation of neural stem cells. Furthermore, Park et al.^[113] harnessed the exceptional electrical conductivity of graphene-based hydrogels to develop artificial muscle constructs.

Despite its exceptional electrical and thermal conductivity and the tunability of its functional properties, graphene is expensive, hydrophobic, and difficult to process. These characteristics,

particularly its high production cost, limit the feasibility of graphene and related materials for cost-effective, large-scale manufacturing. Nonetheless, modified graphenes and/or GOs have remarkable properties that make them highly promising for high-value applications, especially in advanced medical technologies.^[114,115] Finally, modified forms of graphene typically exhibit reduced electrical conductivity and inferior quality compared to pristine graphene. This prompts the search for alternative carbon-based nanomaterials with enhanced and optimized properties.

3.1.1.4. CNTs: Single-Walled or Multiwalled

Since their emergence in the early 1990s, CNTs have garnered considerable attention across multiple disciplines, including physics,^[116] chemistry,^[117,118] and materials science.^[119] This interest stems from their extraordinary mechanical strength, electrical conductivity, and chemical stability, which are attributes derived from their unique cylindrical structure composed of sp^2 -hybridized carbon atoms arranged in a hexagonal lattice.

CNTs are primarily classified into two types: single-walled CNTs (SWCNTs) and multiwalled CNTs (MWCNTs). SWCNTs are composed of a single graphene sheet rolled into a seamless cylinder with diameters typically between 1 and 2 nm. In contrast, MWCNTs consist of multiple concentric graphene cylinders, with diameters ranging from 2 to 25 nanometers and interlayer spacings close to 0.34 nm—similar to that of graphite.^[120,121] The electronic properties of CNTs are heavily influenced by their structural parameters such as chirality and diameter. Depending on these features, CNTs can act as either metallic or semiconducting materials.^[122] SWCNTs, in particular, demonstrate well-defined quantum behaviors, including single-electron tunneling and Coulomb blockade at low temperatures.^[123–126] CNTs exhibit several nanoscale advantages, such as high surface-to-volume ratios and quantum confinement effects, which contribute to enhanced sensitivity, selectivity, and electrochemical responsiveness.^[127] These properties have facilitated their widespread use in diverse technological applications, including chemical sensors,^[128–131] catalyst supports,^[132] energy devices,^[127,128] and nanoelectronics.^[124] CNTs offer enhanced electron transfer capabilities as electrochemical electrodes^[133] at the electrode–solution interface^[134–136] and outperform traditional materials in conductivity and stability. Their use in electrochemical transducers improves the sensitivity and signal-to-noise ratio of enzyme electrodes,^[137,138] immunosensors,^[139] and nucleic acid sensors.^[140,141] They are well-suited for biosensors, supporting protein immobilization while preserving biological activity.^[142]

CNT-modified electrodes are particularly well-suited for amperometric biosensors that utilize oxidase and dehydrogenase enzymes. These electrodes enhance the electrochemical detection of key analytes such as nicotinamide adenine dinucleotide (NADH)^[143–145] and hydrogen peroxide,^[146,147] both critical in enzymatic sensing schemes. CNTs not only amplify biocatalytic reactions but also facilitate electron transfer when configured as vertically aligned arrays or “CNT forests.” In these

arrangements, CNTs function as nanoscale molecular wires that promote efficient electron communication between enzymes and electrode surfaces.^[148] Due to their unique structural and electronic properties, CNTs have found widespread application in electrochemical sensors,^[133,149] actuators,^[133] and energy storage technologies.^[150] The electrical behavior of CNTs is fundamentally linked to the nature of their charge carriers. In graphite, the delocalized π -electrons within its hexagonal lattice facilitate semiconducting properties. CNTs, however, exhibit distinct electrical characteristics arising from their π -bonding and quasi-1D structure. This structure confines electron motion, resulting in a quantized density of states and the emergence of a bandgap at the Fermi level.

The bandgap in CNTs is inversely related to their diameter; larger-diameter CNTs exhibit smaller bandgaps, supporting low-scattering, long-range electron transport over micrometer distances, provided that structural defects are minimal.^[151] Electron transfer through CNTs is governed by the availability of conducting states, each capable of supporting quantum conductance by transmitting spin-paired electrons or holes. However, charge carrier reflection at the CNT-metal contact interface can impede this transfer, limiting access to conducting states. This interaction introduces a voltage drop at the contact point, although not along the nanotube itself.^[151]

A particularly remarkable feature of CNTs is their ability to carry extremely high current densities of up to 10^{10} A cm², exceeding that of conventional metals by three to four orders of magnitude.^[152] Combined with their exceptional mechanical robustness and chemical stability, these traits position CNTs as strong candidates for replacing silicon in future electronic devices. They are also integral to the development of nanoelectromechanical systems (NEMs).^[153] Owing to their metallic or semiconducting character, individual CNTs can function as both transistors and interconnects. This duality is especially valuable in the context of modern semiconductor technology, where traditional silicon-based devices are approaching their physical scaling limits.^[154] CNT-based electronics represent a promising pathway toward enhanced device miniaturization and performance optimization, potentially enabling the continued advancement of Moore's law through novel circuit architectures.^[155]

3.1.1.5. GC

GC, also known as vitreous carbon or glass-like carbon, is one of the most commonly used electrodes for electrochemical sensing applications. It can be produced through low temperature pyrolysis at a temperature range of 300 °C–1200 °C, resulting in sp^2 -hybridized carbon arranged in a hexagonal pattern.^[156] GC is a low-ordered, hard, nongraphitizing, and isotropic carbon. It can be prepared through carbonization of phenol formaldehyde resin.^[157] Additionally, it can be derived from various renewable sources like the processing of the lignocellulosic biomass of plants and their residues such as corncobs, rice husks, and sugarcane bagasse.^[158] Other precursors of vegetable origin used for GC production are sucrose, cellulose, and tannin.^[156]

GC is available in several forms, including cellular GC (CGC), monolithic GC (MGC), and reticulated GC (RGC). Among these, RGC is the most commonly used electrode in electrochemical sensors owing a good electrical conductivity, high open porosity, and large surface area.^[159] In electrode fabrication, GC is embedded within an insulating tube and can function as either a rotating disk or stationary electrode. GC electrodes are frequently modified and enhanced with nanomaterial electrocatalysts, carbon materials, metals, metal oxides, conducting polymers, or their hybrids. Such modifications can improve the electrode's reliability, selectivity, and sensitivity for the detection of pollutants.^[160] In addition to using modifying materials, many techniques may be adopted to modify GC electrodes, including mechanical polishing, chemical treatment, and plasma treatment. These methods are commonly applied for surface cleaning and activation by introducing functional groups onto the GC surface, thereby increasing its active area and improving its electroanalytical performance.^[156]

GCEs exhibit low electrical resistivity and can be fabricated in the form of films, sheets, or powders.^[160] Their low density is attributed to the presence of numerous structural pores.^[156] Moreover, GC displays excellent impermeability to gases and liquids, along with high stability and favorable thermal ($0.7\text{--}4\text{ W m}^{-1}\text{ K}^{-1}$) and electrical conductivities ($10^1\text{--}10^4\text{ S m}^{-1}$).^[161] Particularly, GC exhibits high mechanical stability upon contact with radicals contrarily to graphite which degrades and displays a clearer degradation profile when it exposed to them.^[162] Many studies have shown that electrochemical sensors made from unmodified GC offer rapid and cost-effective analytical tools.^[156]

However, these electrodes have certain limitations. They are prone to surface oxidation and fouling. Most sp^2 -hybridized carbon surfaces can react electrochemically with dissolved oxygen in aqueous solutions, leading to the formation of phenols and lactones.^[59] The resulting surface charge can significantly impact adsorption, electron transfer kinetics, and electrocatalysis.^[163] Furthermore, GC has a potential window of 1.5 to 2 V, which is wider than that of metal electrodes, yet still too limited for covering a broad range of redox reactions.^[67] Despite these challenges, the application of GC electrodes in electrochemical sensors has been widely explored for detecting various chemical compounds and biomolecules, particularly in pharmaceutical applications, food safety, clinical analysis, and environmental monitoring and remediation.^[164]

3.1.1.6. Carbon Fiber (CF)

CF, also known as graphite fiber, is typically described as a material made up of long chains of carbon atoms, containing $\approx 90\%$ or more carbon. The fibers have diameters ranging from 5 to 10 micrometers.^[165] CFs are extremely strong and possess excellent chemical resistance properties. They are widely used in many applications due to their remarkable properties, including small size, high strength, sensitivity, flexibility, and thermal

resistance.^[165] Sometimes, to minimize costs, engineers use a combination of glass and carbon to reinforce the fibers.^[166]

CFs can be used as integrated sensors eliminating the need for external sensors, thus making the system easier to produce and more cost-effective. Moreover, they can serve as temperature and humidity sensors in multifunctional composites. They also possess high surface area and excellent electrical properties that make them suitable as sensors for additional applications such as resistive heating, electromagnetic shielding, energy storage, and health monitoring.^[167] CF microelectrodes can be used in vivo to measure rapid changes in neurotransmitters and investigate fundamental neurobiological processes and the effect of pharmaceutical substances including drugs of abuse.^[168] CFs are also applied as electrochemical sensors for measuring sweat cortisol.^[169]

Over the last few years, there has been an increasing demand for the use of recycled CF as a sustainable sensor in various applications to reduce the material costs due to their unique mechanical strength and recyclability. This sustainable electrode is prepared using composite materials consisting of recycled CF and epoxy resin. They are attractive for structural health monitoring because of their radiolucency. Nowadays, the use of recycled CF, being highly conductive and durable, is very promising for integrating sensing into passive structures and systems (such as automobile parts and assistive technologies) and supporting the improvement of new types of capacitive-based sensors such as touch, pressure, proximity, and soil moisture sensors. This sustainable sensor could be a good alternative in applications where intrinsic sensing is desired.^[170]

3.1.1.7. Carbon Black (CB)

CB is a form of amorphous carbon that has a very fine particle size of tens of nanometers. It exhibits a high degree of structural order because of the layering of graphene-like sheets (sp^2 hybridized carbon atoms) on the external surface, combined with edge-plane-like defects associated with sp^3 hybridized carbon atoms.^[171] CB is typically fabricated via incomplete combustion or thermal decomposition of heavy aromatic oils or hydrocarbons under specific conditions.^[172] The size and the physicochemical properties of CBs are often process-dependent.

Like many other carbon-based materials, CB is widely used due to its high surface area to volume ratio, high chemical and thermal stability, relatively good electrical conductivity, and low cost.^[173] In the sensory field, CB is often formulated into a paste or composite material, where it serves as a conductive matrix when mixed with a binder or a high dielectric permittivity material such as polymers and ionic liquids.^[174] This formulation enhances both mechanical stability and electrochemical performance, making it suitable for many electrode applications. CB-based electrodes are typically used in low-cost, disposable sensors or as supporting materials for immobilizing enzymes, redox mediators, or nanoparticles to enhance sensitivity and selectivity in the electrochemical measurements.^[175,176]

3.1.1.8. Biochar-Based Material

Biochar is a carbon-rich material produced from the pyrolysis of biomass (thermal decomposition of organic waste in the absence of oxygen (anaerobic conditions)). The increasing demand for renewable and sustainable carbon sources makes biochar an alternative replacement for traditional graphite and activated carbon materials.^[177] They are organic-derived materials which offer various advantageous especially its porosity, high surface area, stability, and eco-friendly and sustainable source. This material is typically used for energy storage devices but is also used for the formulation of electrochemical sensors and (bio)sensors.^[178] More precisely, the high porosity structure makes biochar materials ideal for adsorption process (to be used as adsorbent materials). This characteristic offers biochar a good physicochemical properties idea for the electrochemical analysis of pollutants. Additionally, biochar exhibits a catalyst property making it a good candidate for the construction of sensors and (bio)sensors. Biochar have been reported in the literature for the detection of heavy metals, pesticides, veterinary drug, environmental pollutants, and biomolecules.^[175,177,178]

Finally, carbon-based materials have emerged as highly effective components in the design of electrochemical sensors and (bio)sensors. These materials serve as an ideal choice for immobilizing biomolecules promoting efficient electron transfer and enabling flexible sensors designs making them suitable for a wide range of applications (Table 1).

3.1.2. CPs

CPs are essential for the development of electrochemical sensors. Due to their unique electrical and physicochemical properties, they are used in many applications such as biosensors, environmental monitoring, and chemical detection. In this part, we focus on four CPs that are commonly used in electrochemical sensing platform: 1) polyaniline (PANI), 2) polypyrrole (PPy), 3) poly(3,4-ethylenedioxythiophene) (PEDOT), and 4) poly(3,4-ethylenedioxythiophene): polystyrene sulfonate (PEDOT: PSS) (Figure 5). The key properties, advantages, disadvantages, and

main applications in electrochemical sensors are presented and discussed in Table 2.

3.1.2.1. PANI

PANI is commonly used in electrochemical sensors for its favorable properties: first, PANI is electrochemically active due to its redox activity (i.e., it has tunable redox states), which facilitates electron transfer processes essential for sensor functionality.^[179] PANI shows moderate conductivity that can be enhanced with protonic acid. However, the nanostructured forms of PANI (nanotubes, nanofibers) have a high surface area which exhibits high surface area, which increases the sensor sensitivity.^[180] Additionally, PANI demonstrates good chemical stability in neutral and acidic environments and good resistance to moderate temperatures.^[181] PANI can also be functionalized with specific receptors or nanoparticles, leading to a high selectivity. Its biocompatibility is suitable for biosensor applications, especially when combined with enzymes or antibodies.

PANI is essentially used to functionalize the WE in electrochemical sensors. It facilitates the detection of analytes by providing a conductive surface for electron transfer. PANI is widely available, inexpensive, easy to prepare, and can be easily doped with many materials to enhance its properties for specific applications. However, PANI lacks mechanical strength, and its conductivity can be affected by environmental conditions such as humidity or pH.

Many fabrication techniques can be used to incorporate PANI: 1) electrodeposition—^[182] direct deposition on electrode surfaces through electrochemical polymerization, forming a conductive film; 2) screen printing—^[183] inks are printed onto substrates to create sensor patterns; 3) inkjet printing—^[184] precise deposition of PANI onto substrates, creating complex sensor designs; and 4) electrospinning—^[185] producing PANI nanofibers enhancing surface area and sensitivity of sensors.

PANI-based sensors are mainly used in pH sensing but many other applications were also reported including hydrogen peroxide detection important in biomedical diagnostics, metal ion detection useful in environmental monitoring, and organic compound detection in chemical analysis.^[186]

Table 1. Comparative table summarizing electrochemical properties of carbon-based materials (e.g., solvent window, background current, electron transfer kinetics, detection limits, and sensor applications) for each material class.

Material	Solvent window ^{a)} (vs Ad/AgCl)	Background current ^{b)}	Electron transfer kinetics	Detection limits ^{c)}	Applications
BDD	Very wide ≈ -1.5 to $\approx +2.5$ V	Very low	Fast	nM to pM	[34,65,319]
Graphite	Moderate ≈ -1.0 to $\approx +1.0$ V	Moderate	Moderate	μ M to nM	[4,29,89,175]
Graphene	Wide ≈ -1.5 to $\approx +1.5$ V	Low to moderate	Fast	nM to pM	[104–106,320–323]
CNTs	Moderate ≈ -1.0 to $\approx +1.5$ V	Low	Vert fast	nM to pM	[116,117,154,189,324]
GC	Moderate ≈ -1.0 to $\approx +1.2$ V	Low	Fast	μ M to nM	[102,325,326]
CF	Moderate ≈ -0.8 to $\approx +1.0$ V	Moderate	Moderate	μ M to nM	[94,167,327,328]
CB	Narrow ≈ -0.5 to $\approx +0.8$ V	Moderate to high	Moderate to slow	μ M range	[171,175,176]
Biochar-based materials	Variable ≈ -0.5 to $\approx +1.0$ V	Moderate to high	Variable (often slow)	Variable but in μ M range	[178,323,329,330]

^{a)}Solvent window values are approx. and depend on electrolyte and electrode treatment. ^{b)}Background current is qualitative. ^{c)}Detection limits vary widely depending on analyte, sensor configuration, and measurement technique.

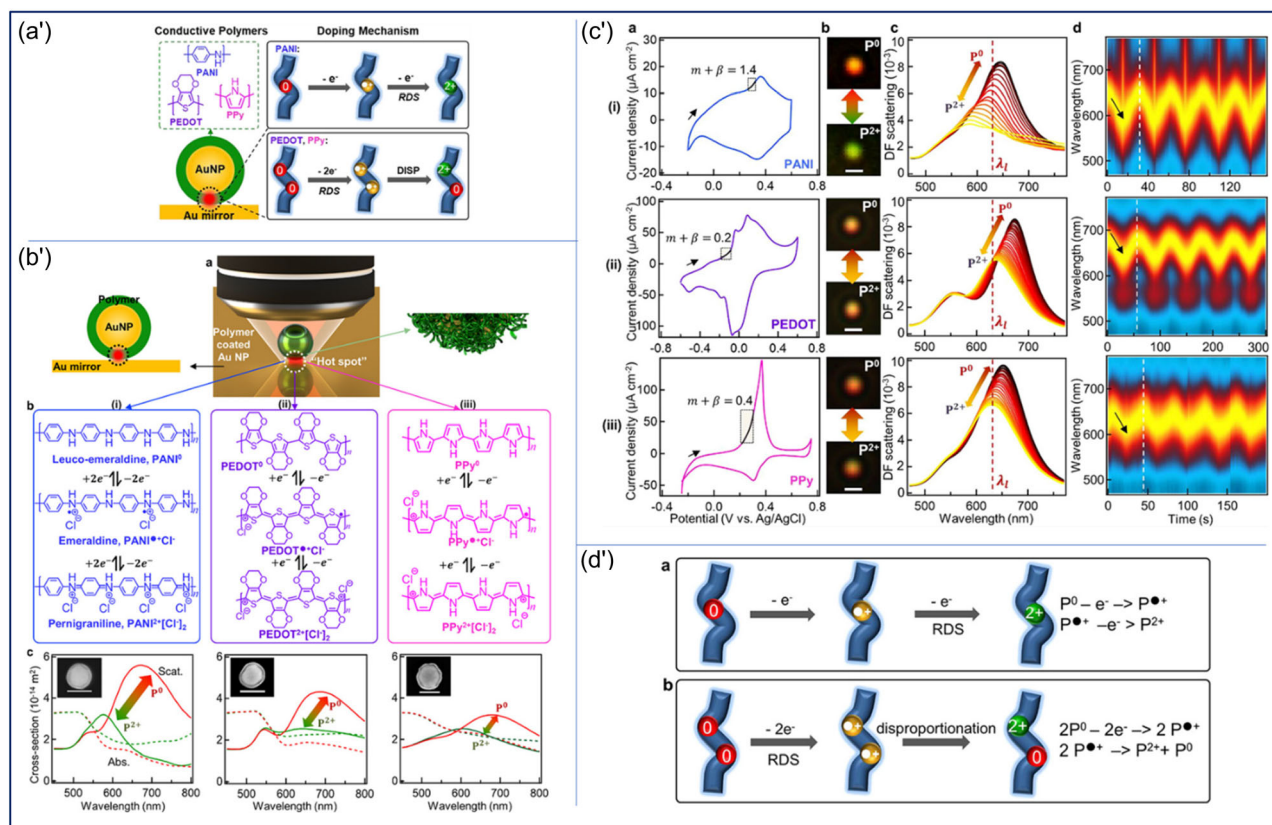


Figure 5. a') CPs used for the construction of electrochemical sensors. b') CP redox in plasmonic hotspots. a) Schematic of electrochromic nanoparticle-on-mirror (eNPoM). Each gold nanoparticle (Au NP) is coated with (i) PANI (Au@PANI), (ii) poly-3,4-ethylenedioxythiophene (Au@PEDOT), or (iii) PPy (Au@PPy). b) Redox polymer reactions. c) Finite-difference time-domain (FDTD) simulated absorption (dashed) and scattering (line) spectra of individual eNPoM optical switching of eNPoMs. (i) Au@PANI, (ii) Au@PEDOT, and (iii) Au@PPy. c') (a) CV with apparent transfer coefficients ($m + \beta$), (b) Dark-field (DF) scattering images, scale bar 1 μm . (c) DF scattering spectra of single eNPoMs as potential applied. Dashed $\lambda_l = 633 \text{ nm}$ is laser wavelength used for surface-enhanced Raman spectroscopy (SERS, see below). d) Time-series normalized DF scattering spectra from single eNPoMs over 5 cycles. d') Mechanism of CP redox: Two pathways for oxidative doping: (a) consecutive two-step electron transfer by losing two electrons step-wise. (b) Disproportionation-type electron transfer by losing one electron; then, two P^+ rapidly combine to form a P^{2+} without extra electron transfers.^[318]

Table 2. Most important properties of CPs commonly used in electrochemical sensors.

Property	PANI	PPy	PEDOT	PEDOT/PSS
Conductivity	Moderate, enhanced by doping	Moderate, enhanced by doping	High enhanced by doping	Very High
Redox activity	Excellent; multiple oxidation states	Good	Moderate limited to certain oxidation states	Moderate
Catalytic activity	Moderate enhanced with composites	Moderate enhanced with composites	Low not typically catalytic	Low not typically catalytic
Surface area	High especially in nanofiber form	High especially in nanotube form	Moderate enhanced with composites	Moderate enhanced with composites
Chemical stability	Good stability in acidic and neutral environments	Good stability in basic and neutral environments	Excellent stability in a wide range of pH	Excellent stability in a wide range of pH
Thermal stability	Moderate degradation above 150 °C	Moderate degradation above 150 °C	High stable up to 200 °C	High stable up to 200 °C
Biocompatibility	Moderate improved with composites	High	High	High
Cost and availability	Low cost, available	Low cost, available	Moderate; more expensive than PANI and PPy	Moderate; more expensive than PANI and PPy
Fabrications methods	Screen printing, inkjet printing, electrodeposition	Screen printing, inkjet printing, electrodeposition	Screen printing, inkjet printing, electrodeposition	Spin coating, Screen printing, inkjet printing, electrodeposition
Main applications	pH sensors, ammonia gas sensors, glucose sensors	Glucose sensors, dopamine sensors, drug detection	Glucose biosensors, neurotransmitter detection, organic matter detection	Glucose biosensors, neurotransmitter detection, organic matter detection

3.1.2.2. PPy

Compared to PANI, PPy has similar properties. PPy is electrochemically active due to its reversible redox activity; it facilitates electron transfer processes used for electrochemical sensing applications. PPy exhibits moderate electrical conductivity due to its conjugated π -electron system. Its conductivity can be enhanced through doping with anions such as *p*-toluene sulfonate (*p*-TSA) or poly(styrene sulfonate) (PSS).^[187] The electrochemical polymerization of PPy can be controlled to form nanostructures, which increase the surface area and enhancing the sensitivity of sensors. Additionally, PPy demonstrates good chemical stability in neutral and basic environments; however, its thermal stability is limited under high temperatures.^[188]

Like PANI, PPy is essentially used to functionalize the WE in electrochemical sensors. It facilitates the detection of analytes by providing a conductive surface for electron transfer. PPy can be integrated into electrochemical sensors using various techniques: 1) electrodeposition—^[189] direct deposition onto electrode surfaces through electrochemical polymerization, forming a conductive film; 2) screen printing—^[190] PPy inks are printed onto substrates to create sensor patterns, suitable for mass production; and 3) inkjet printing—^[191] allows for precise deposition of PPy onto substrates, enabling the creation of complex sensor designs.

PPy-based sensors are also used in various applications^[192,193] including pH sensing, hydrogen peroxide detection, metal ion detection, and organic compound detection.

3.1.2.3. PEDOT

PEDOT exhibits high conductivity, often enhanced by doping agents like PSS making it suitable for various electronic applications. PEDOT demonstrates reversible redox behavior, facilitating electron transfer processes essential for electrochemical sensing. The electrochemical polymerization of PEDOT can be controlled to form nanostructures, increasing the surface area and enhancing the sensitivity of sensors.^[194] PEDOT shows very good chemical stability in various environments; however, its thermal stability can be limited, which may affect sensor performance under high-temperature conditions. By functionalizing PEDOT with specific receptors or nanoparticles, sensors can achieve high selectivity towards target analytes.^[195] PEDOT has good biocompatibility suitable for biosensor applications, especially when combined with bioreceptors like enzymes or antibodies.

PEDOT is also used to functionalize the WE in electrochemical sensors. PEDOT is relatively inexpensive compared to other conducting polymers. It is easy to prepare it and can be doped with many materials to enhance its properties for many applications. However, PEDOT conductivity can be affected by environmental factors like humidity and pH, and it lacks mechanical strength, necessitating the use of composites for integrity.

PEDOT can be integrated into electrochemical chips using the same techniques as PPy: electrodeposition,^[196]

screen-printing,^[197] and inkjet printing.^[198] PEDOT-based sensors are used in identical applications like PANI and PPy-based sensors.^[199,200]

3.1.2.4. PEDOT: PSS

PEDOT: PSS is a conducting polymer that exhibits high electrical conductivity, typically ranging from 10 to 100 S cm⁻¹. Its conductivity can be enhanced by doping with polar solvents such as DMSO or surfactants such as ethylene glycol, which can increase the conductivity up to hundreds or even thousands S cm⁻¹. PEDOT: PSS demonstrates reversible electrochemical behavior, facilitating electron transfer processes essential for electrochemical sensing applications.^[201] The electrochemical polymerization of PEDOT: PSS can be controlled to form nanostructures, which increase the surface area and enhance the sensitivity of sensors.^[202] It shows good chemical stability in various environments; however, its thermal stability can be limited, which may affect sensor performance under high-temperature conditions. Similarly, to PEDOT, PEDOT PSS is highly selective and biocompatible.^[203]

PEDOT: PSS is primarily used to functionalize the WE in electrochemical sensors. It facilitates the detection of analytes by providing a conductive surface for electron transfer. The reference and CE typically remain unmodified to maintain stable potential and current during measurements.

Comparing to PANI, PPy, and PEDOT, PEDOT: PSS is distinguished by its ability to be fully processed in water. Thus, making PEDOT: PSS cost-effective, environmentally friendly, and safe in contact with biological agents (enzymes, antibodies, etc.) It can be integrated into electrochemical chips using various techniques:^[204]

1) electrodeposition—PEDOT: PSS is directly deposited onto electrode surfaces through electrochemical polymerization, forming a conductive film; 2) screen printing—PEDOT: PSS inks are printed onto substrates to create sensor patterns, suitable for mass production; 3) inkjet printing—allows for precise deposition of PEDOT: PSS onto substrates, enabling the creation of intricate sensor designs; and 4) spin coating—an aqueous dispersion of the polymer blend is spin coated onto a substrate using speeds between 1000 to 5000 rpm to control film thickness. Then, the film is dried to remove water and enhance conductivity and film uniformity.

PEDOT: PSS-based sensors are used in various applications.^[205,206] PEDOT: PSS's protonation and deprotonation properties make it suitable for pH detection. Additionally, PEDOT: PSS's redox activity enables the detection of hydrogen peroxide, which is important in biomedical diagnostics. It can be also used for metal ion detection, functionalized PEDOT: PSS sensors can detect metal ions, useful in environmental monitoring. And PEDOT: PSS sensors can be tailored to detect specific organic compounds, supporting chemical analysis. PEDOT: PSS is widely available and cost-effective, making it an attractive material for sensor development. Its synthesis methods are well-established, and it can be produced in various forms to suit different sensor applications.

To conclude, all these CPs play a major role in sensor signal generation due to their electrical properties, chemical sensitivity, and capacity to undergo reversible doping/dedoping processes.

They can transform chemical interactions into measurable electrical signals, making them ideal for use in electrochemical and chemi-resistive sensors. For example, PANI-based sensors have been used for ammonia detection through changes in conductivity upon gas exposure, while PEDOT: PSS has been integrated into biosensing platforms for glucose and dopamine sensing via enzymatic reactions or redox interactions. The versatility of CPs enables their application in wearable devices, environmental monitoring, and biomedical diagnostics, where sensitivity and flexibility are paramount.

3.2. Modifying Materials

3.2.1. Metal-Based Materials and Alloys

3.2.1.1. Gold

Gold-based materials including nanoparticles (AuNPs), nanorods, thin films, and composite structures are widely used in electrochemical sensing due to their distinctive physicochemical properties. These include excellent electrical conductivity, high chemical and physical stability, biocompatibility, and large surface to volume ratio.^[207] Gold-based materials can also be functionalized with organic, inorganic, or biological molecules (enzymes, proteins, DNA, etc.) enabling versatile sensor design.^[23] Additionally, gold has tunable optical and electronic properties allowing control of their size, shape, and morphology.^[208] These properties enhance the sensitivity and selectivity of (bio)sensors especially when using as a WE. These characteristics make gold-based materials ideal as transducers in biosensors.^[209] The binding of a biological element, e.g., an antibody, enzyme, or aptamer to its target analyte can cause detectable changes in the transducers' physicochemical properties that produce a measurable signal.^[210]

Gold-based (bio)sensors have been developed for the detection of a wide variety of analytes, including metal, ions, proteins, nucleic acids, microorganism, and disease biomarkers. These systems are used in medical diagnostics (e.g., cancer, COVID-19, and Parkinson's disease) as well as agriculture and environmental applications.^[211,212] Common sensing strategies include DNA-functionalized gold nanostructure-based colorimetric assays, electrochemical immunosensors, microfluidic-integrated systems, and interdigitated electrode-based sensors. Fluorescence and electrochemiluminescence approaches using gold nanomaterials have also shown promise for the detection of small biomolecules such as cholesterol,^[213] hematin,^[214] and uric acid.^[215]

On the other hand, integration of gold-based materials onto the electrode surfaces can be achieved via different methods: 1) electrodeposition, 2) drop casting, and 3) ink-based fabrication (3D printing, inkjet printing, screen-printing, etc.).^[216] Despite their many advantages, the relatively high cost of gold deposition remains a limiting factor for widespread use in low-cost or large-scale applications.

3.2.1.2. Platinum

Platinum-based materials are the most used sensing material for electrochemical sensors, because of their good catalytic activity, electronic conductivity, biocompatibility, and chemical stability.^[217] They have unique properties such as their surface effect, volume effect, quantum size effect, and macroscopic quantum tunneling effect and can be synthesized by various methods to obtain materials with different physical and chemical properties.^[218] Platinum possesses many of the aforementioned characteristics and has been used as a CE material for decades due to its inertness.^[219] Many synthesis methods of platinum-based materials exist where the chemical synthesis method accurately controls the size and shape and affords strong surface chemical versatility and high yield.^[218] Platinum-based material can be also used as a WE and in three ways as electrochemical catalysts to accelerate the reaction of enzymes, as sensor electrodes to enhance electron transfer, and to modify the electrode surface with an enzyme to fix the substrate and maintain enzyme activity.

Furthermore, Pt nanoparticle-based biosensors have potential applications in the agricultural and environment sectors, for example, detection of pesticide residues particularly phosphate ester compounds in organophosphorus pesticides, with low detection limit, high analyte selectivity, and high universality using Pt nanomaterials as amperometric acetylcholinesterase biosensors (AChE).^[220] These AChE amperometric biosensors are sensitive, accessible, and cost-effective compared with the traditional methods of chromatography and spectrometry used for detecting the environmental pollution caused by organophosphorus pesticides.^[218]

Moreover, Pt NP-based biosensors play an important role in the field of medicine. Pt-based materials have exceptional cellular biocompatibility^[221] like gold-based materials; however, PtNPs have a greater cytotoxic effect,^[222] which could be used in cancer therapy since platinum ions can interact with DNA inhibiting its replication.^[223] Platinum-based materials have also shown antibacterial activity against Gram-positive and Gram-negative pathogens, similar to that of silver-based materials, although silver present greater toxicity than PtNPs, which limits their clinical use.^[224] Platinum-based biosensors can be used to efficiently and rapidly test various chemical components in body fluids, such as glucose detection, as well as graphene platinum-based hydrogen peroxide biosensors for reactive oxygen species (ROS) detection. PtNP are also highly sensitive for detecting lactic acid, uric acid (its high accumulation can cause shock, dehydration, and kidney diseases), and glutamic acid (its concentration imbalances can cause psychiatric disorders such as Alzheimer's disease and Parkinson's disease), along with many carcinogens.^[218]

Platinum nanomaterials and biosensors based on platinum are also widely used in the food sector^[225] (for the detection of lactic acid that exists in many foods, beverages, and fermented products). Moreover, platinum-coated magnetic beads/palladium nanoparticles (Pt/PdNP) are used for the detection of Salmonella in food and fecal samples.

Moreover, platinum is mainly prepared by the electrodeposition method onto the 3D electrodes^[226] which can significantly

reduce the overpotential of H_2O_2 and enhance the electron transfer rate between the electrode and the Pt nanoparticles. Platinum-based materials present many advantages compared to their disadvantages, which are only noticed in few anticancer drugs in biomedical sector due to their dose-limiting toxicity.^[227]

3.2.1.3. Silver

Silver-based materials have been widely applied in electrochemistry because of their ease of preparation, high conductivity, and relative stability.^[228] Silver is the best electrical conductor of all metals because it has the lowest electrical resistance. Silver wire coated with silver chloride is relatively insensitive to changes in temperature.^[229] Silver electrodes are considered among the popular REs due to their simple fabrication.^[230] Its excellent electrical properties, including a reversible reaction and the ability to provide stable and reproducible potential, make silver ideal for applications such as REs in electrochemical experiments and pH electrodes.^[229] Moreover, the role of silver as an alloying additive is highlighted when reducing the overpotential of the oxygen evolution reaction on lead anodes used in the electrowinning of zinc.^[231] In addition, the influence of silver on the electrochemical and corrosion properties of Pb–Ca–Sn–Al grid alloy in sulfuric acid solutions at room temperature was investigated by potentiodynamic and potentiostatic methods.^[232] However, AgCl electrodes face poor biocompatibility which makes this material challenging in contact with biological agents.

On other hand, silver nanoparticles are among the most frequently used NPs because of their promising catalytic, electronic, chemical, magnetic, optical, and electrochemical properties.^[233] For the synthesis of AgNPs, chemical and biological methods are applied by reducing Ag^+ ions to metallic silver (Ag^0) using suitable reducing agents. Conventional chemical reducing reagents are often toxic and nonbiocompatible compounds. Thus, biological reducing agents have been tested as biodegradable and benign compounds for the synthesis of AgNPs. Plants, fungi, and bacteria can be used in the synthesis of AgNPs.^[233] Electrodes with modified AgNPs are known for their increased surface area and superior conductivity compared with unmodified electrodes.

Accordingly, silver-based materials have been extensively applied in industrial, biomedical, and environmental applications. In industrial applications, they are used as catalysts for reducing bicarbonate to formate and entering in oxygen reduction reactions. By this process, they can be applied for energy storage, the production of synthetic fuels, and clean energy technologies.^[233] Recently, AgNPs have gained great attention in the biomedical sector. They have been utilized as key chemotherapeutic drugs to treat various types of cancers, such as pancreatic, prostate, breast, ovarian, and blood cancers. In addition, Ag-based electrochemical sensors are considered a promising tool for determining various organic pollutants in water, thus controlling environmental pollution and subsequent water treatment. Like gold, silver-based materials are expensive compared with other common metals. However, they have various uses and are mainly used for antimicrobial and anticancer therapy and applied in the promotion of wound repair and

bone healing or as the vaccine adjuvants, antidiabetic agent, and biosensors.^[234] Despite the use of Ag-based materials in biomedical applications, dental materials, biosensors, biotechnology, and agriculture, doping Ag materials with small quantities of a second metal may enhance its efficiency while maintaining biocompatible levels.^[235] Gold and platinum are excellent candidates as dopants because they have a greater electrochemical potential than silver and higher biocompatibility.^[235]

3.2.1.4. Nickel

Nickel nanoparticles (NiNPs) have harvested significant attention in the field of electrochemical sensing due to their unique physical and chemical properties, cost-effectiveness, and versatile applications.^[236] Their properties, feasibility in sensor applications, and advantages and disadvantages as well as recent developments in electrode designs incorporating NiNPs are discussed in the following.

NiNPs exhibit several properties that make them suitable for electrochemical sensor applications such as high surface area, excellent electrocatalytic activity, chemical stability, and magnetic properties.^[237] The nanoscale size of NiNPs provides a large surface-to-volume ratio, enhancing the electroactive area while facilitating electron transfer.^[238] They can catalyze various redox reactions, making them effective in detecting analytes like glucose, hydrogen peroxide, and cholesterol while demonstrating good stability under alkaline conditions which is beneficial for certain sensing applications.^[239] In addition, the intrinsic magnetism of the metal allows for easy separation and manipulation in sensor fabrication processes.^[240] The integration of NiNPs into electrochemical sensors has been extensively explored, demonstrating their feasibility across various applications. Porous nickel electrodes, for example, have been developed for nonenzymatic glucose detection, demonstrating high sensitivity and stability,^[241] while for insulin sensing, NiNP-modified electrodes have been fabricated for ultrasensitive detection of insulin achieving low detection limits and wide dynamic ranges^[242] (Figure 6).

The incorporation of NiNPs into electrode materials offers various advantages. Being more abundant and less expensive than noble metals such as gold, nickel offers cost-effectiveness thus making NiNP-based sensors more economically feasible.^[243] The ease of fabrication and versatility of NiNPs enable them to be synthesized and integrated into sensors using various methods such as electrodeposition and chemical reduction^[236] while they can be combined with other nanomaterials such as graphene or CNTs to create composite electrodes with customized properties while leading to improved sensitivity in detecting various analytes.^[244,245] Despite their benefits, the use of NiNPs in electrochemical sensors poses certain challenges. They are susceptible to oxidation under certain conditions, thus affecting their stability and performance.^[246] Their properties may be altered further due to the formation of aggregates, which can reduce their effective surface area.^[247] Moreover, while the metal is generally considered biocompatible, certain concerns may surface regarding its potential toxicity in certain applications.^[248]

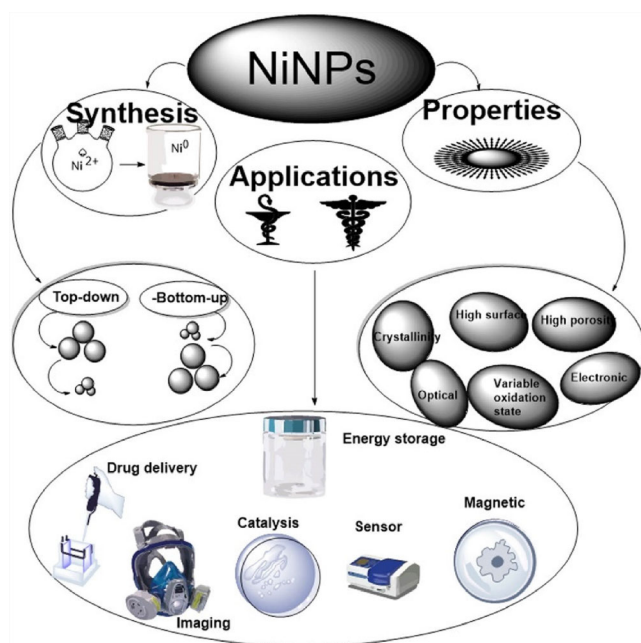


Figure 6. NiNPs synthesis, properties, and applications.^[236]

Recent studies have focused on innovative electrode designs incorporating NiNPs to overcome existing challenges and enhance sensor performance. Electrodes modified with NiNPs and graphene have shown improved electrocatalytic activity and sensitivity in detecting analytes like diclofenac,^[249] while nickel-based metal-organic frameworks (Ni-MOFs) have been integrated into flexible electrodes for glucose sensing, demonstrating high sensitivity and mechanical stability.^[250] In addition, NiNPs have been utilized to modify SPEs, which offer a low-cost and scalable approach for sensor fabrication.^[251]

Overall, and in terms of activity, stability, and selectivity, NiNP-modified electrodes exhibit high catalytic activity, facilitating redox reactions and improving sensor responsiveness while demonstrating good stability under certain conditions where surface modifications and composite formations are often used to enhance their long-term stability. Through surface functionalization and the use of selective recognition elements, NiNP-based sensors can achieve high selectivity for various target analytes. NiNPs have significantly contributed to the advancement of electrochemical sensors, offering enhancements in sensitivity, selectivity, and cost-effectiveness. While challenges such as oxidation and aggregation remain, ongoing research into innovative electrode designs and composite materials continues to address these issues. The unique properties of NiNPs position them as valuable components in the development of next-generation electrochemical sensors for a wide range of applications, from medical diagnostics to environmental monitoring.

3.2.1.5. Palladium

Owing to their high biocompatibility, exceptional catalytic properties, large surface/volume ratio, electrical conductivity,

structural robustness, and stability, PdNPs have shown their significant role in materials for the field of electrochemical sensing and are widely used in sensor development.^[252] They exhibit high catalytic activity, facilitating redox reactions essential for electrochemical sensing.^[253] Regarding electrical conductivity, they perform well in ensuring efficient electron transfer, thus enhancing sensor sensitivity.^[254] Moreover, PdNPs are chemically stable in preserving their functionality over extended periods and under various environmental conditions.^[255] Such exceptional physical and chemical properties have rendered them appropriate for various applications such as the detection of biomolecules, environmental pollutants, and industrial chemicals.^[252,256,257] Although they have achieved exceptional performance in glucose and reducing gas detection, they suffer from low sensitivity and weak selectivity in various applications.^[252]

PdNPs incorporation into electrochemical sensors has been extensively studied while showing their feasibility across various applications.^[258] More specifically, PdNPs have been exploited in nonenzymatic sensors for the detection of hydrogen peroxide while exhibiting high sensitivity and selectivity.^[259] They have also been entrapped in nanoporous gold wires to develop sensitive detection of dopamine revealing their potential in neurological diagnostics.^[260,261] In addition, PdNP-modified carbon spheres combined with molybdenum disulfide have been used for the electrochemical detection of quercetin, a flavonoid with antioxidant properties.^[262]

Much like other nanoparticles sensors, PdNPs demonstrate advantages, disadvantages, and certain challenges in sensor design. Enhanced sensitivity, improved selectivity, and rapid electron transfer were among the advantages that PdNPs have provided in electrochemical sensor applications. Through the high surface area and catalytic activity of PdNPs, their incorporation provided amplification of electrochemical signals thus allowing for the detection of analytes at lower concentrations.^[252,259,263] On the other hand, and while taking dopamine as an example of target analyte, its selectivity was improved by the functionalization of PdNPs with specific recognition elements that ensured selective binding to target analytes while reducing interference from other substances.^[261] This was achieved by embedding PdNPs with 4-aminophenol functional nitrogen-doped graphene quantum dots (PdNPs/4AP N-GQDs) which produced greater selectivity towards dopamine via strong hydrogen bonds in the presence of other interfering substances such as ascorbic acid and uric acid, which oxidize within the same potential window as dopamine. Moreover, PdNPs were shown to facilitate faster electron transfer rates, leading to strong and quicker sensor responses.^[264]

Despite these benefits, the use of PdNPs have also yielded certain disadvantages and challenges such as high cost, formation of aggregates, and surface fouling. Being a precious metal, large scale production of PdNP-based sensors tends to be cost-prohibitive.^[265] Aside from this, PdNPs had the tendency to form aggregates which lead to the reduction in their effective surface area that altered their properties necessitating the frequent utilization of stabilizing agents or surface modifications to maintain dispersion.^[266,267] Moreover, nonspecific adsorption of proteins or

other molecules onto them was found to interfere with their sensing accuracy.^[253] To overcome these challenges, recent studies have focused on innovative electrode designs to enhance sensor performance. For example, PdNP-modified carbon spheres were developed by combining the metal nanoparticles with carbon spheres and molybdenum disulfide which led to the development of core-shell composites for the sensitive detection of quercetin, an important flavonoid that used to treat colitis and gastric ulcers and is a promising anticancer agent.^[262] Another study has targeted the selectivity and high sensitivity for dopamine detection by entrapping PdNPs in self-supporting nanoporous gold material. The nanocomposite material was found to be more sensitive to dopamine than any of its closest competitor in the literature.^[261] Further integration of PdNPs with graphene quantum dots and ZnFe₂O₄ has produced electrochemical sensors with enhanced detection of glucose that were proven to be robust, reusable, versatile, and sensitive.^[268] PdNPs have significantly contributed to the development and advancement of electrochemical sensors, offering enhancements in sensitivity, selectivity, and stability.

Although challenges such as cost and aggregate formation remain, ongoing research into advanced electrode designs and composite materials continues to tackle these issues. The unique properties of PdNPs make them as valuable components in the development of next-generation electrochemical sensors for a wide range of applications, from environmental monitoring to medical diagnostics.

3.2.1.6. Metal Oxide-Doped Nanoparticles (MOXNPs)

MOXNPs have emerged as fundamental components in the development of advanced electrochemical sensors. The exceptional physical and chemical properties such as high catalytic activity, surface-to-volume ratio, and tunable electronic properties make them suitable for detecting various analytes ranging from environmental pollutants to biomolecules.^[269] This section explores the properties, feasibility, advantages, challenges, and recent advancements in the application of MOXNPs in electrochemical sensor technologies.

The combination of nanostructures and the modification of sensing materials through the integration of metal oxide with other metal particles or other elements have a considerable impact on sensor performance by enhancing their properties, performance, and functional capabilities. This approach helps in the creation of a larger active surface area, increasing the adsorption of targets onto the sensing material, and increasing the sensor selectivity towards the analyte, while decreasing the interference from others. The nanoscale dimensions of MOXNPs provide a large surface area that facilitates the interaction with target analytes while improving sensor sensitivity.^[270,271] Despite these advantageous, many metal oxides suffer from intrinsically low electrical conductivity. This can lower the electron transfer and reduces sensitivity of electrochemical sensors.^[272] This limitation makes MOXNPs less adequate as standalone electrode materials. To overcome this limitation,

MOXNPs are often combined with highly conductive materials such as graphene, CNTs, or metal nanoparticles.^[273] These composites improve the overall performance of the sensing platform. In most cases, metal oxides serve as functional surface modifier or catalytic components rather than primary electron conductor matrix.^[274] Certain metal oxides, such as SnO₂ and ZnO, demonstrated semiconducting behavior which can be exploited to adjust electron transfer processes in sensors.^[275] MOXNPs can also catalyze oxidation reduction reactions, thus enabling the detection of analytes without the need for additional mediators or enzymes.^[276,277] Under various environmental conditions, many metal oxides were found to be chemically stable which ensures the longevity and durability of the sensors.^[278,279] The integration of MOXNPs into electrochemical sensors along with other materials such as graphene, CNTs, and metal nanoparticles has been extensively studied, expressing their feasibility across various applications such as environmental monitoring, biomedical diagnostics, and food safety.^[269] MOXNPs in environmental analysis namely SnO₂ and ZnO have been used in the detection of pollutants such as heavy metals and nitrates in water samples.^[280] Iron oxide nanoparticles have been used in biosensors for biomedical diagnostics purposes in detecting biomolecules such as dopamine and glucose.^[281] As for food safety, copper oxide nanoparticles have revealed their potential in detecting contaminants such as metronidazole in foodstuffs.^[282,283]

In electrochemical sensing, MOXNPs have been shown to offer various advantages such as robust fabrication using thin and/or thick film technologies, simple operation, cost-effective production, selectivity, and rapid response.^[284] Due to their increased surface area and catalytic characteristics, MOXNPs provide more surface sites for interaction of analytes and hence higher efficiency in transferring electrons resulting in enhanced sensitivity and amplified electrochemical responses.^[285,286] Through attaching specific ligands to MOXNPs, the functionalized metal oxide nanoparticles may be designed to interact with specific targets thus increasing selectivity. This tailoring allows the production of specific and precise sensors with minimized interferences from other substances as well as the synthesis in various shapes and sizes, enabling customization of sensor properties to suit specific applications may be realized.^[269,287,288] Moreover, the effective electron transfer facilitated by these materials resulted in quicker sensor responses thus improving the process of electrochemical detection which potentially contributes to the development of highly efficient sensing platforms for diverse applications.^[269,289] While MOXNPs offer various advantages in electrochemical sensors, their utilization also produces several hurdles and challenges that must be overcome for optimal performance.^[269] Such disadvantages may include stability and reproducibility issues, functionalization and/or modification for optimal performance, and interferences as well as the complexity in the development of portable on-site sensors. The synthesis and immobilization of MOXNPs have been found to be difficult to control especially with varying sizes and shapes hence producing different topology between sensors.^[290] Furthermore, the properties of such composite nanoparticles may be affected under varying environmental conditions thus leading to reproducibility

issues.^[269,290] In terms of specificity and selectivity, MOXNP-based sensors faced a challenge when analyzing complex environments with various analytes.^[285]

In addition, other components in the matrix of real samples were found to interfere with the biomolecular interaction thus compromising the accuracy of the sensor.^[291] To reach optimum performance, MOXNPs-based sensors often require functionalization, which adds complexity to the fabrication process and thus necessitates the need for higher energy consumption as well as higher production costs.^[292,293] Additionally, the development of on-site portable sensors that are based on MOXNPs and that yield high sensitivity and reproducibility still remains a challenge.^[291] To address limitations and challenges in existing sensor technologies and to improve their functionality in several aspects, recent research has been greatly focused on developing new designs that incorporate MOXNPs. Enhancing stability and sensitivity as well as overall sensor efficiency are among the studies that were carried by controlling the unique properties of the MOXNPs discussed earlier.^[269,290,294,295] Besides gold and platinum, MOXNPs such as ZnO and Fe₃O₄ were used to significantly increase sensor functionality by providing higher surface area and catalytic activity that increased sensitivity while achieving quicker responses times.^[269,294–296]

Research work is engaging in studying various electrode designs that incorporate MOXNPs that include nanoporous, self-healing, flexible/stretchable, and multimodal and multispectral electrode sensors. The nanoporous nature of the electrodes offers a high surface area which in combination with MOXNPs improved the sensing ability of the electrodes.^[295–298] To extend the lifespan and increase the stability of sensors, self-healing electrodes that utilize OD materials such as graphene quantum dots were explored as filler components that can repair damage and restore functionality.^[299] To add more flexibility especially in medical applications or wearable technologies, innovative designs such as the Kirigami-inspired structures were used to maintain the functionality of the sensors under moving and stretchable conditions.^[299–301] Applications of such improved sensors may

have great potential in a wide range field that may include industrial and environmental monitoring as well as health care.

The unique characteristics of MOXNPs place them as valuable components in the development of the next-generation electrochemical sensors for a broad range of applications. Such nanoparticles have significantly contributed to the advancement of electrochemical sensors by offering enhancements in sensitivity, selectivity, and stability. Ongoing research into such innovative electrode designs continues to address challenges and issues in an attempt to refine them into fully practical, efficient, and reliable sensors (Table 3).

3.2.2. Redox Dyes and Mediators

Redox-active dyes and mediators such as Prussian blue (PB), methylene blue (MB), and Meldola blue (MB') are particularly interesting in the development of electrochemical sensors. These compounds are electrochemically active and undergo reversible redox reactions. This makes them effective electron mediators in a variety of electrochemical sensing applications.^[302] They facilitate indirect electron transfer between nonconductive biological elements such as enzymes and aptamers and the electrode. This enhances signal strength and electrode sensitivity. They incorporation typically aims at reducing overpotentials, improving selectivity and enabling detection of electrochemically inactive species at low potentials preventing thus interference problem. They can be easily formulated into printable inks or electrochemically deposit onto electrode surfaces enabling efficient electron transfer and allowing for batch fabrication processes (Figure 7).

3.2.2.1. PB

Several studies in the literature have reported the use of PB as a redox mediator in electrochemical sensors. This mediator is particularly attractive due to its ability to shift the oxidation or reduction potentials of target analytes. PB functions as an artificial

Table 3. Comparative properties of nanoparticles in electrochemical sensors.

Nanoparticle type	Signal enhancement	Catalytic activity	Selectivity	Stability	Applications
Gold (AuNPs)	High	Moderate to high	Good with thiol-functionalization	Excellent (chemically inert)	Biosensors, immunosensors, glucose detection ^[139,207,211]
Silver (AgNPs)	High	High (especially for redox reactions)	Moderate (prone to oxidation)	Moderate (oxidation susceptible)	Antibacterial sensors, heavy metal detection ^[331,332]
Platinum (PtNPs)	High	Very high	Moderate	Very high	Hydrogen peroxide, enzymatic biosensors ^[333,334]
Copper (CuNPs)	High	High (oxidation catalysts)	Low to moderate	Poor to moderate (oxidation)	Pesticide detection, nitrite sensors ^[335,336]
Zinc oxide (ZnO NPs)	Moderate	Moderate	Moderate to high (after functionalization)	Good	Environmental pollutants, enzymatic sensors ^[337–339]
Iron oxide (Fe ₃ O ₄ NPs)	Moderate	Moderate	High (bioconjugation capability)	High (especially in composites)	Biosensors, magnetic separation-assisted sensing ^[340–342]
Carbon-based NPs	High	Low (unless modified)	High (easy functionalization)	Excellent	Multiplexed detection, wearable sensors ^[189,324,343]

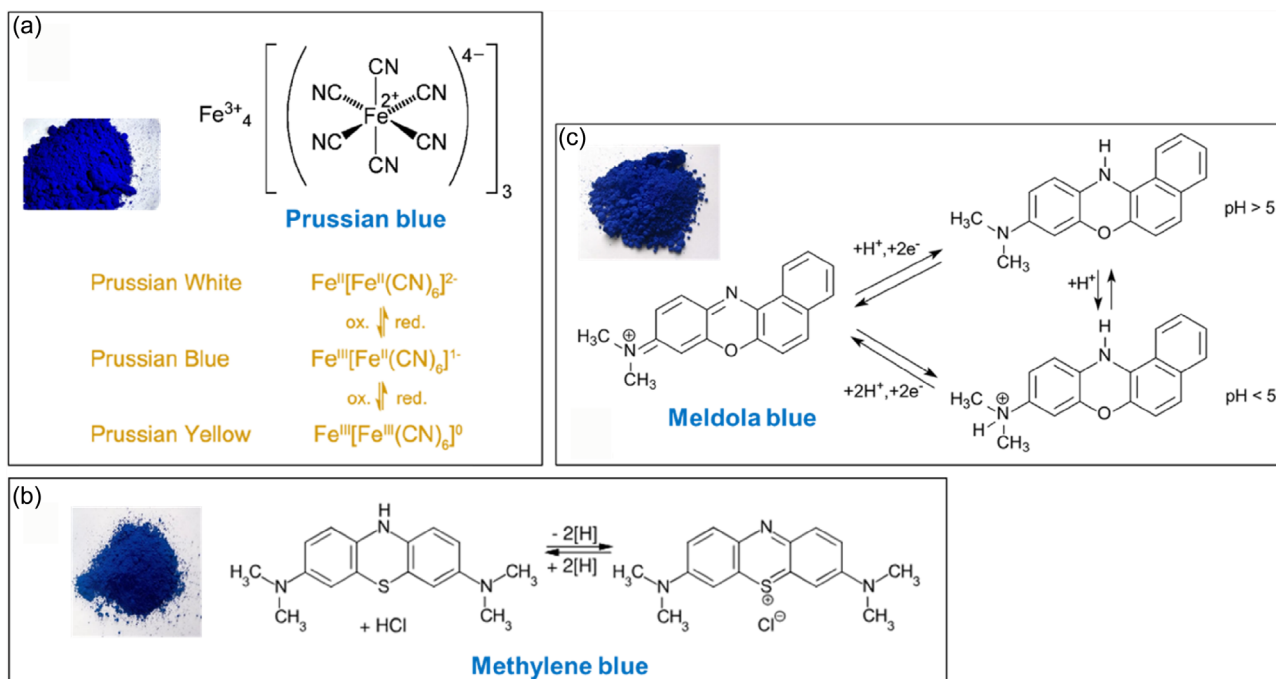


Figure 7. Chemical structures and redox mechanism of: a) PB, b) MB, and c) MB'.

peroxidase, which catalyze the reduction of H_2O_2 at low potentials (between -0.1 and 0 V vs. Ag/AgCl). This significantly minimizes interference from electroactive species present notably in biological samples. This redox cycling involves the interconversion of PB states ($\text{Fe}^{3+}/\text{Fe}^{2+}$) and its reduced form Prussian white facilitating continuous electron flow from the enzymatic reaction product to the electrode.^[303]

Rodrigo Vieira Blasques et al. developed a disposable electrochemical sensor incorporating PB as a mediator, capable of simultaneously detecting multiple relevant biological analytes.^[304] The sensor was designed to analyze both enzymatic analyte such as glucose using glucose oxidase that produces hydrogen peroxide as a product and nonenzymatic analytes such as hydrogen peroxide or uric acid.

The use of PB enabled a significant lowering of the detection potential thereby minimizing interference from nontarget species. Electrodes were fabricated by depositing a graphite conductive ink (GNP) onto a plastic substrate, followed by the electrodeposition of a PB film. Surface characterization using scanning electron microscopy (SEM) revealed a uniform, porous PB layer, while energy-dispersive X-ray spectroscopy and infrared spectroscopy (FTIR) confirmed the chemical composition and functional groups. Cyclic voltammetry (CV) characterization indicated stable redox properties and increased electroactive surface area. These characteristics highlight PB's suitability for biomedical sensing applications, especially in low-cost, portable devices for point-of-care-diagnostics.

A similar design incorporating PB into a multiparametric sensing platform was reported by Chmayssem et al.^[33] The authors developed a multiparametric (bio)sensor platform connected to a microfluidic system designed to enable real-time monitoring

of key cellular stress metabolites including pH, lactate, hydrogen peroxide (H_2O_2), and nitric oxide (NO).^[305] Platform fabrication was based on screen-printing technology including the PB ink printing on specific electrodes (lactate and hydrogen peroxide) followed by a drop-casting deposition of the enzymatic membrane for the lactate sensor.

3.2.2.2. MB

MB is well-known redox mediator that undergoes reversible redox electron transfer reactions. Its low redox potentials and electrochemical reversibility make it a convenient electron shuttle in various electrochemical systems. MB redox cycling occurs via a reversible two-electron process involving its oxidized (MB^+) and reduced (leuco-MB) forms.^[306] This enables it to mediate electron transfer efficiently in systems where direct oxidation of target analytes is difficult. In DNA and aptamer-based sensors, MB often acts as both a structural probe and redox label, with its current signal modulated by conformational changes upon target binding.^[307,308] Many studies were reported in the literature developing functional sensors based on MB for detecting various organic solvents and volatile gases,^[309] biological species,^[310,311] environmental pollution,^[312] etc.^[313]

These sensors typically involve the fabrication of thin films often by deposition techniques followed by the characterization of their spectroscopic and electrochemical responses. The results reveal high sensitivity, good reproducibility, and satisfactory selectivity for specific analytes. However, challenges remain, particularly concerning the long-term chemical stability and performance in complex or variable environments. Future development

efforts aim to improve material stability and enable portable, field-deployable sensor platforms.^[309]

In a separate and promising application, researchers have developed an electrochemical aptasensor for the detection of *Pseudomonas aeruginosa*, a common pathogen responsible for hospital-acquired infections. The sensor utilizes a specific DNA aptamer immobilized on a modified carbon-glass electrode, with MB serving as the redox indicator. MB interacts with the aptamer via π - π stacking, allowing changes in its redox signal to report on target binding.^[311] Electrochemical techniques including CV, differential pulse voltammetry, and electrochemical impedance spectroscopy (EIS) were used for detection. The sensor demonstrated a limit of detection (LOD) of ≈ 1.5 CFU mL⁻¹ and a linear detection range from 10¹ to 10⁵ CFU mL⁻¹. The total assay time was under 60 min, making it suitable for rapid diagnostics. The device showed excellent specificity, successfully distinguishing *P. aeruginosa* from nontarget bacteria such as *Escherichia coli* and *Staphylococcus aureus*.

3.2.2.3. MB'

MB' is particularly interesting in electrocatalytic detection of NADH, a common cofactor in enzymatic reactions. NADH's direct oxidation at bare electrodes typically occurs at high potentials (around +0.6 V vs. Ag/AgCl) which risks fouling the surface. MB' overcomes this limitation by catalyzing NADH oxidation at significantly lower potentials (at -0.05 V vs. Ag/AgCl), thus preventing electrode stability and enhancing detection sensitivity.^[314]

Several studies have focused on developing innovative electrochemical sensors for the detection of NADH using a carbon nanofiber-modified electrode and MB' as the redox mediator. The method used by Ana Maria Titoiu et al. enables the enzymatic detection of benzaldehyde by coupling it with bacterial aldehyde dehydrogenase (ALDH), which produces NADH as a reaction product.^[315] The generated NADH is then detected electrochemically through its redox interaction with immobilized MB'. Despite these advantages, the system's long-term stability is compromised by the gradual desorption of MB' from the electrode surface. This highlights the need for improved strategies for mediator immobilization to enhance durability.

Another study conducted by Chmayssem et al. aimed to study the evolution of the cell culture media composition by studying the influences of the cell culture media on the response of electrochemical sensors.^[34] One of the studied sensors for the indirect detection of nitric oxide (NO) used MB' to monitor nitrite/nitrate, the oxidation byproducts of NO.

Furthermore, a sensitive electrochemical sensor for the detection of NADH has been developed using MB' as a redox mediator adsorbed onto a carbon paste electrode.^[316] NADH, a key cofactor in numerous enzymatic reactions, is challenging to detect directly due to its high oxidation potential and the risk of electrode surface passivation. This study addresses these limitations by using MB', a redox-active dye known for its electrocatalytic properties,

to lower the oxidation potential of NADH and enhance signal stability. The sensor was fabricated by adsorbing MB onto a carbon paste electrode, and electrochemical performance was evaluated using CV and amperometry. The results demonstrate that the modified electrode facilitates the electrocatalytic oxidation of NADH at a substantially lower potential (≈ -0.05 V vs. Ag/AgCl) compared to the $\approx +0.6$ V required on an unmodified electrode. This shift is critical for minimizing signal interference and preventing electrode degradation. To address stability concerns, more robust immobilization strategies for MB' such as covalent attachment are being considered. Additionally, optimizing electrode architecture and coupling the system with microfluidic platforms may enable miniaturization and on-site diagnostics.

4. Conclusion

The design and material selection of electrochemical cells play an important role in determining the performance and sensitivity of electrochemical (bio)sensors. With the growing global demands for rapid diagnostics, environmental monitoring, and food safety assurance, the demand for cost-effective and reliable sensors becomes more critical. Advances in electrode configurations, innovative materials, and integrated detection strategies continue the drive progress of the sensory field. Future research should focus on connecting the laboratory innovation with real-world applications closely to ensure that technologies remain accessible and impactful in improving the quality of life worldwide.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: electrochemical sensors · electrochemistry · electrode design · electrode materials

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